Participant Handbook

Foundations of Clinical Research · Level 1: From Clinical Question to Publication

Provider: Neudata · **#ClearDataClearImpact**

Welcome

Every day you make clinical decisions that research could improve. Over eight weeks, with one online session every Saturday evening (19:30 to 22:30, Vietnam time), this course takes you through the whole research process: from a question at the bedside, to a sound study design, to analysing data, to writing and publishing the paper.

You do not need any research or statistics background. We go slowly, with pictures, one running example and a lot of short polls and questions in the chat. The course is fully online and you join from home. There are no groups or breakout rooms: the whole class works through every exercise together, with the instructor. You will never be asked to calculate anything beyond a simple 2×2 table, and you do not need any software: the trainers run the analyses, and you learn to read and write them up.

The course at a glance

Week	Session (Saturdays, 19:30 to 22:30, online)	Theme	You build (homework)
1	1	**ASK:** from clinical problem to research question	Proposal: topic, PICO, question, aims
2	2	**FIND & READ:** searching, managing references, reading a paper in 15 minutes	Proposal: search, 5 references, problem statement, working title
3	3	**DESIGN:** choosing the right study	Proposal: design, setting, population
4	4	**BIAS & EFFECT:** bias, confounding, RR/OR/NNT, diagnostic tests, forest plots	Proposal: exposure, outcomes, bias plan
5	5	**DATA:** CRFs, data management, describing data, *meeting the course dataset*	Proposal: variables, CRF, data plan · Manuscript: Table 1 understood
6	6	**ANALYSE:** CIs, p-values, tests, regression, survival, missing data, *analysing the course dataset*	Proposal: sample size, analysis plan · Manuscript: results interpreted
7	7	**PROTECT & WRITE:** ethics, registration, integrity, and writing Methods and Results	Proposal: ethics, feasibility, publication plan · Manuscript: Methods + Results

Week	Session (Saturdays, 19:30 to 22:30, online)	Theme	You build (homework)
8	8	**PUBLISH:** Introduction, Discussion, abstract, journals, submission, peer review and presentations	Manuscript: complete draft · Proposal submitted (about 5 volunteers present)

Each session lasts 3 hours with a 15-minute break. It has a short recap, two teaching blocks with frequent "check your understanding" moments (a quick poll or an answer in the chat), and a proposal-builder segment in which the instructor shows the next part of the proposal on the running example. You then write the same part for your own question at home. Each session also has extra slides at the end for anyone who wants to read further at home.

The running example

One clinical topic travels with us: **fluids in septic shock**. In Weeks 1 to 4 we ask, search and design the question, and look at two real papers (a randomised trial and a cohort study). In Weeks 5 to 8 we read the results of FLUID-ICU, a realistic *simulated* dataset of 620 ICU patients, and you write them up as a paper.

Your two products (individual)

Everyone works on their own products. There are no groups.

1. **A research proposal on your own question**, from your own department, built one part at a time as homework and submitted in Session 8 (written proposal + 6 slides). About 5 volunteers also present their proposal in Session 8 (6 minutes + 2 minutes of feedback); presenting is optional. See `Assignment/proposal_brief.md`.
2. **A draft manuscript from the course dataset.** You choose ONE of the five FLUID-ICU questions, Q1 to Q5 (fluid balance and mortality, kidney injury, ventilator-free days, lactate and missing data, ICU length of stay). We give you the results in a results pack, and you write the paper. The instructor models Methods and Results in Session 7 and the rest in Session 8; you write your own as homework. After the course, you review one classmate's draft (assigned by the instructor) and reply to the review of yours like a real author. See `Assignment/manuscript_brief.md` and `Assignment/peer_review_instructions.md`.

Between sessions (one week; about 30 to 60 minutes)

After	Do this
Session 1	Draft your topic, PICO, research question, FINER check, aim and objectives; think of synonyms for your PICO terms
Session 2	Finish Proposal Part 1 and the literature matrix; read the abstract of Paper 2
Session 4	Finish Proposal Part 2
Session 5	Finish Proposal Part 3 (variables, CRF, data plan); look at the five results packs (Q1 to Q5)
Session 6	Finish Proposal Part 3; choose your manuscript question (Q1 to Q5) and read its results pack; read Papers 1 and 2; start your 6 slides

After	Do this
Session 7	Finish Proposal Part 4 and your 6 slides; draft the Methods + Results for your question
Session 8	Finish and submit the manuscript; review one classmate's draft within 1 week; response letter within 2 weeks

What you need

A computer (or tablet) with a stable internet connection, camera and microphone (a headset helps), the meeting link sent before Session 1, and a quiet place for 3 hours. A pen and a notebook. No statistics software to install. Exercise sheets and other handouts are sent as PDFs before each session. Participants who have done the free Neudata course *Clinical Data Analysis in R* may re-run the course analyses with the scripts in `Data/R/`.

Your toolkit

Resource	File
Research proposal template	`Templates/Research_Proposal_Template.docx`
Study protocol template	`Templates/Study_Protocol_Template.docx`
Informed consent template	`Templates/Informed_Consent_Template.docx`
CRF + data dictionary template	`Templates/CRF_Template.docx`
Literature review worksheet	`Templates/Literature_Review_Worksheet.docx`
Statistical analysis plan template (+ worked example `Data/SAP_FLUID-ICU.md`)	`Templates/Statistical_Analysis_Plan_Template.docx`
Manuscript (IMRaD) template	`Templates/Manuscript_Template_IMRaD.docx`
Cover letter and response-to-reviewers templates	`Templates/Cover_Letter_Template.docx`, `Templates/Response_to_Reviewers_Template.docx`
Conference abstract & poster guide	`Templates/Conference_Abstract_Template.docx`
Critical appraisal checklist (10 questions)	`Resources/Critical_Appraisal_Checklist.pdf`
Risk-of-bias tool chooser	`Resources/Risk_of_Bias_Tool_Chooser.pdf`
Writing & submission checklist	`Resources/Writing_and_Submission_Checklist.pdf`
Sample ethics application, sample grant concept note (fictional)	`Resources/`
2×2 and sample-size calculators	`Resources/tools/`
Course dataset, data dictionary, results packs	`Data/`
English–Vietnamese glossary, further reading	`References/`

The quiz

In Session 1 and again at the end of Session 8 you answer the same 24 questions online (a link is posted in the chat). You give your full name so that your two attempts can be compared. Only the course team sees names and scores, and you do not see the answers until after Session 8. The quiz measures what the *course* achieved, not you.

After the course

Take your proposal to the ethics committee and keep improving your manuscript. This course is Level 1. If you want to run analyses yourself, Level 2 is a hands-on course in R (prerequisite: this course and the free *Clinical Data Analysis in R* course). Level 3 offers advanced, specialised methods on request (`Course_Levels/`).

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